

Empirical Study of How Personality, Team Processes and Task Characteristics Relate to Satisfaction and Software Quality

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ABSTRACT

This paper analyses the relationships between *personality*, *team processes*, *task characteristics*, *product quality* and *satisfaction* in software development teams. The data analysed here were gathered from a sample of 35 teams of students (105 participants) from a Spanish university. These teams applied an adaptation of an agile methodology, eXtreme Programming (XP), to develop the same software product.

We found that the teams with the *highest job satisfaction* are precisely the ones whose members score highest for the personality factors *agreeableness* and *conscientiousness*. The *satisfaction* levels are also higher when the members can decide how to develop and organize their work. On the other hand, the level of *satisfaction* and *cohesion* drops the more *conflict* there is between the team members. Finally, the teams exhibit a significant positive correlation between the personality factor *extroversion* and software product *quality*.

Categories and Subject Descriptors

D.2.9 [Software Engineering]: Management – *programming teams*.

General Terms

Management, Experimentation.

Keywords: Personality factors; Team processes; Task characteristics; Software quality; Satisfaction

1. EXPERIMENTAL DESIGN

This paper describes an experimental study that analysed the relationships between the big five personality factors (*neuroticism*, *extroversion*, *openness to experience*, *agreeableness* and *conscientiousness*) at team level, the task characteristics (*autonomy* and *interdependency*) and the team processes (*cohesion* and *conflict*) for the purpose of evaluating their

relationship to the quality of the developed software and the mean satisfaction of the team members in an academic setting with undergraduate students developing software using an adaptation of the agile XP method. It follows in the research footsteps of Barry et al. [1] that related personality to individual and group performance. This empirical study run fits the description of a quasi-experiment [3]. Quasi-experiments are run when the subjects cannot be randomly assigned to an experimental condition or, alternatively, a treatment cannot be randomly assigned to a group. Statistical tests, like correlations and regressions, are applied to the data collected, as applicable. This study follows the guidelines for reporting empirical research in software engineering [6] as closely as possible.

1.1 Variables and Instrumentation

Input and process measures (*personality*, *cohesion*, *conflict*, *interdependency*, *autonomy* and *satisfaction*) were collected at the individual level and aggregated to the mean level. The team's professor graded the deliverables, and then we assessed the *quality* of the software product.

1. *Personality*. The NEO FFI test was used to evaluate the team member personality [4]. This test is composed of 60 questions that measure the five personality factors: neuroticism, extroversion, openness to experience, agreeableness and conscientiousness.
2. *Cohesion* and *conflict*. 13-item questionnaire for evaluating team processes (cohesion and conflict) taken from the Gross Cohesion Questionnaire [9] and an Intragroup Conflict Scale [7]. Conflict is divided into *social conflict* and *task conflict*.
3. *Interdependency* and *autonomy*. The questionnaire used to determine the task characteristics (interdependency and autonomy) is composed of 12 items. Task interdependency is determined following van der Vegt et al. [10], whereas task autonomy is measured using Molleman's questionnaire [8].
4. *Satisfaction*. Gladstein's questionnaire was used to evaluate satisfaction. Specifically, team satisfaction was evaluated using three items from Gladstein [5], indicating how satisfied people claim to be with their team mates, working together, etc. This is the response variable used in the regression analyses.
5. *Software Quality*. To measure the quality of the software product, the professor analysed the code and project documents. Also the professor recorded the progress made in the exercises to quantitatively document each team members' participation. We calculated the weighted grade on a scale from 0 to 10 points according to the following formula:

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Grade = (((Modularization * 2 + Testability * 2 + Functionality * 2 + Reusability * 2 + Style * 2) / 4) * 0.8) + ((Participation * 10 / 4) * 0.2)).

All the personality factors, as well as the task characteristics, team processes and satisfaction are evaluated on Likert scales. Students were told that there are no right or wrong answers and that their responses should be accurate and truthful, taking into account the scoring system (1 = totally disagree, 5 = totally agree).

1.2 Participants and Subjects

The participants were second-year computing students at Universidad Autónoma de Madrid's Higher Technical School of Computing during the 2004/05 academic year. These students were taking the Data Structures and Algorithms course. The project set was to design and implement a distributed library management system. Development took place throughout the whole semester and professors acted as users. The size of the resulting products ranged from 87.15 to 130 adjusted function points.

A total of 105 students participated in this empirical study, of which 83 were male (80%) and 22 were female (20%). Of the students, 75% (79 students) were aged less than 21 years and 25% (26 students) were from 21 to 30 years of age. The students were divided into 35 three-member teams who worked together to develop the software. Each three-member development team is a subject of this quasi-experiment. These teams were formed at random and their members were blind to the quasi-experimental conditions. Teams used a common agile methodology (an adaptation of XP [2] to the academic environment) and the C programming language.

1.3 Data Collection Procedure

All the questionnaires used were anonymised. The teams were identified by a team number and the participants by a team member number (1, 2 or 3). This subject and participant identification was the same for all the questionnaires at each measurement time. Information was gathered by means of measurements taken before, during and after project development.

Before refers to when the project kicks off, the time during which the teams are being set up, but no team work has yet been done. At this time the NEO FFI *personality* test was handed out and collected in. *During* is the period, according to estimated project time, it takes to complete 45% of the development project (week 11). The *cohesion* and *conflict* questionnaires were handed out and collected in. Finally, *after* is, according to estimated project time, when the project is 95% complete (week 23) just before project development is finished. The questionnaires handed out at this point measure the task characteristics (*interdependency* and *autonomy*) and the *satisfaction* of the teams after their team work.

1.4 Quasi-Experiment Validity

Before running the quasi-experiment, we identified three main internal validity threats:

1. *Process conformance*. This was mitigated by allocating two professors to every 60 students. These professors were responsible for monitoring project development.
2. *Knowledge of XP and development experience*. This was mitigated by forming teams at random and organizing a training session on the XP methodology applied to project development.

3. *Software project to be developed*. To prevent this threat, all the teams undertook the same project.

For results to be generalized to an industrial setting, special-purpose quasi-experimental studies at software developer organizations would have to be planned and designed.

2. DATA AND RESULTS ANALYSIS

The following statistical techniques were applied for data and results analysis:

1. Analysis of the correlations of the personality factors, task characteristics (*autonomy* and *interdependency*), team processes (*cohesion* and *conflict*), *satisfaction* and software product *quality*.
2. Regression between *satisfaction* and personality factors, social *conflict*, task *conflict*, *cohesion* and task *interdependency* and *autonomy*.

2.1 Correlations of Team Factors

The initial analysis of correlations between *personality factors*, task *autonomy* and *interdependency*, team processes (*cohesion* and *conflict*) and *satisfaction* revealed some positive and significant associations (see Table 1).

Table 1 shows that there is a positive correlation in the teams between *extroversion* and another two personality factors, *openness* ($r = 0.414$ $p = 0.014$) and *agreeableness* ($r = 0.480$ $p = 0.003$), apart from with the team process, *cohesion* ($r = 0.469$ $p = 0.005$). Similarly, *neuroticism* correlates negatively with *conscientiousness* ($r = -0.636$ $p = 0.000$) and *extroversion* ($r = -0.336$ $p = 0.049$).

The evaluated *conflict* has a significant negative relationship with *cohesion* ($r = -0.496$ $p = 0.002$) and *satisfaction* ($r = -0.464$ $p = 0.005$) of the software development team.

There is a positive linear relationship between students' mean *satisfaction* and mean team personality scores for *agreeableness* and *conscientiousness* ($r = 0.334$ $p = 0.050$ and $r = 0.341$ $p = 0.045$). *Satisfaction* also correlates positively with task *interdependency* ($r = 0.797$ $p = 0.000$) and *autonomy* ($r = 0.471$ $p = 0.004$). In the practical assignment groups, *interdependency* is related to *autonomy* ($r = 0.599$ $p = 0.000$), *agreeableness* ($r = 0.500$ $p = 0.002$), *conscientiousness* ($r = 0.476$ $p = 0.004$) and *cohesion* ($r = 0.421$ $p = 0.012$).

No significant relationships were found between software *quality* and other evaluated factors like *autonomy*, *interdependency*, *satisfaction*, *conflict* or *cohesion*. Note, however, that all the evaluated teams exhibit a significant correlation between the personality factor *extroversion* and software *quality* ($r = 0.455$ $p = 0.038$). So, we can say that there is a positive relationship between the personality factor *extroversion* and software *quality*.

Analysing the individual performance of the 105 components (theory grade) against team performance, we find that there is a significant relationship ($r = 0.201$ $p = 0.040$).

2.2 Linear Regression of Satisfaction

We built a backward model for regression, in which we entered all the variables and then eliminated them one by one based on the output criteria (lowest absolute value of R^2 ; criterion of $F \geq 0.1$). This produced the model shown in Table 2.

Table 1. Correlation matrix between personality factors, task characteristics, team processes and satisfaction

	NEUROTICISM	EXTROVERSION	OPENNESS	AGREEABLENESS	CONSCIENTIOUSNESS	AUTONOMY	INTERDEPENDENCY	SATISFACTION	CONFLICT	COHESION
NEUROTICISM	1	-0.336*	-0.112	-0.151	-0.636**	-0.148	-0.214	-0.129	-0.034	-0.186
EXTROVERSION		1	0.414*	0.480**	0.256	0.302	0.281	0.153	-0.192	0.469**
OPENNESS			1	0.302	0.052	-0.085	0.090	0.068	0.236	-0.061
AGREEABLENESS				1	0.167	0.503**	0.500**	0.334*	-0.157	0.376*
CONSCIENTIOUSNESS					1	0.216	0.476**	0.341*	0.007	0.261
AUTONOMY						1	0.599**	0.471**	-0.314	0.333
INTERDEPENDENCY							1	0.797**	-0.243	0.421*
SATISFACTION								1	-0.464**	0.294
CONFLICT									1	-0.496**
COHESION										1

p < 0.05* p < 0.01**

Note firstly that we applied a two-step procedure, arriving at a corrected coefficient of determination or correlation R^2 of 0.733 (out of 1). In other words, apart from *social conflict*, the model predicts or explains that degree of variance of the variable *satisfaction* from task *interdependency*, *cohesion* and *task conflict*. A '-' sign in Table 2 indicates a negative relationship. Interestingly, the assessed relationship conflict plays no role in explaining the model.

Table 2. Regression model for satisfaction

	Non-standardized regression coefficients	Standardized regression coefficients
	B	Beta
(Constants)	8.013	
INTERDEPENDENCY	0.521	0.765
COHESION	-0.168	-0.182
TASK CONFLICT	-0.674	-0.380

$R^2 = 0.733$

Of the variables evaluated, the best predictor of satisfaction is *interdependency* with a Beta coefficient = 0.765. Precisely, as we are dealing with small teams carrying out activities that they have to successfully complete very quickly (very short-term product quality), interdependency is indispensable, as, otherwise, there would be a very big sense of frustration at not achieving the objective. This is because in agile software development all the tasks are closely interrelated and must be performed rapidly. Consequently, there are also close and dynamic mutual dependencies between development team members.

Table 3 shows how the linear regression converges with correlations between model variables, where *interdependency* and *satisfaction* correlate, $r = 0.797$ ($p = 0.000$). We also find that *task conflict* ($r = -0.526$ $p = 0.001$) has a closer negative relationship than *social conflict* ($r = -0.354$ $p = 0.037$).

Table 3. Correlations between satisfaction, social conflict, task conflict and interdependency

	SATISFACTION	SOCIAL CONFLICT	TASK CONFLICT	INTERDEPENDENCY
SATISFACTION	1	-0.354*	-0.526**	0.797**
SOCIAL CONFLICT		1	0.637**	-0.178
TASK CONFLICT			1	-0.287
INTERDEPENDENCY				1

p < 0.05* p < 0.01**

3. CONCLUSIONS

On the basis of the results obtained in this quasi-experiment, we can say that the teams most *satisfied* with their job are precisely

the ones that score higher on the personality factors of *agreeableness* and *conscientiousness*. The levels of *satisfaction* are also higher when the team members can decide on how to develop and organize their work. On the contrary, the level of *satisfaction* and *cohesion* drops the greater *task conflict* is among team members. Finally, there is a significant correlation between an average score for the personality factor *extroversion* and evaluated software product *quality*.

Consequently, *extroversion* should be considered as a valid predictor of software quality for developing software following an agile methodology. A high interaction among team members is essential for this development method. All participants could be classed as project managers and are all responsible for the success or failure of the developed product. This way, traits like sociability, talkativeness, communicativeness, affability and openness appear to be conducive to the development of high quality software, as well as to the satisfaction of the development team members.

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